PaisaPath: Final Report

Problem statement: Vanguard, Community Stewardship (financial literacy for Grade 11–12 students)

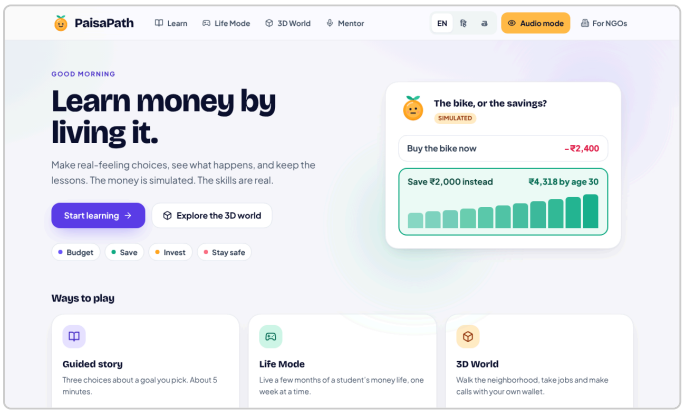
Team No.: ____

1. Problem and our understanding

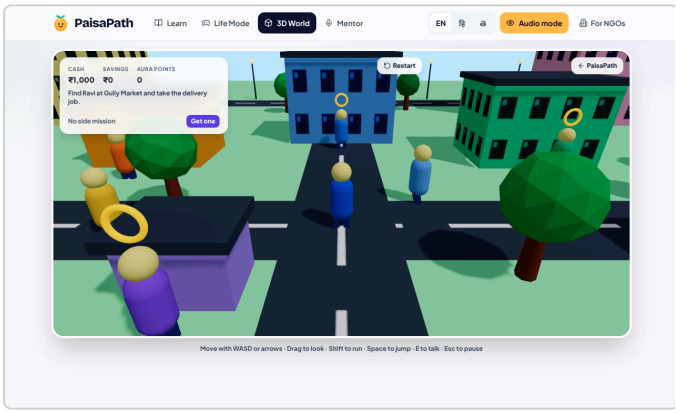
Many Grade 11–12 students in Hyderabad are about to handle their own money for the first time: pocket money, a first part-time income, UPI payments and, very often, scam messages. Classroom finance teaching is abstract, and students with no background feel judged or intimidated. The brief asks for a tool that builds four habits (budget, save, invest wisely, stay safe) in the students' own languages, with one firm guardrail: **educate, never recommend real investments**.

2. Solution overview

PaisaPath is a browser game where students learn money by living it. Every rupee is simulated, but every choice has a visible consequence, a plain-language explanation and a chance to replay. It runs fully offline with no account or API key; AI (Google Gemini) is an optional layer that adds generated scenarios and a conversational mentor. The interface is available in **English, Hindi and Telugu**.



Home: four pillars and the ways to play



3D World: walk the neighbourhood with a live wallet

3. Features

Mode	What the student does	Pillars taught
Guided story (Learn)	Onboarding asks name, language, interests and learning style; a short instinct check sets a starting level; then a goal-based story with three choices and a money-style report at the end.	Budget, save
Life Mode	Lives several months of a student's money life one week at a time across 72 scenarios: pocket-money day, a subscription auto-renewal, a cracked phone screen, a first weekend job, repaying a sibling, KYC/UPI scam messages. Deterministic outcomes, Aura Points, levels, achievements, concept cards with worked examples, and a Future Lab that compares compound growth with fictional numbers only.	All four
3D World	Walks a low-poly Hyderabad-style neighbourhood (ring road, traffic, pedestrians, shops). Takes a delivery job from Ravi at Gully Market, then side missions that follow the wallet: compare bike vs bus fare, take a second shift, pay down debt, simulate a monthly SIP, skip an unnecessary purchase, price-check at the chai cart. A results panel analyses the choices at the end.	Budget, save, invest (simulated)
Mentor	Ask a money question or log spending by voice or text ("I spent 120 on lunch and got 500 from my aunt"). Works offline with a rule-based parser; uses Gemini when available, with model fallback on rate limits.	Budget, stay safe
Audio Mode	Full-screen, spoken-first mode for visually impaired students: Space to talk, R to repeat, opened with Space from anywhere.	Accessibility
NGO dashboard	A facilitator view, shown with sample data, of how a partner NGO would follow a cohort's progress.	Stewardship

4. Approach and architecture

- **Client-only web app.** Vite 5, React 18 and Tailwind 3, with three.js through React Three Fiber for the 3D world. Nothing needs a server, so a school lab or a phone browser can run it from a static build.
- **Pure domain core.** Game rules live in plain JavaScript modules separate from the UI: `src/lifemode/domain` (game state, ledger, progression, replay, report, Future Lab) and `src/neighborhood` (economy, missions, shops, traffic, pedestrians, movement, controls, results analysis). The UI only renders state and dispatches choices, which made the rules unit-testable and deterministic.
- **Ledger-based money.** Every rupee movement is a ledger entry (cash, savings, debt), so balances can never drift and every outcome can be explained back to the student.
- **Content as data.** Scenarios, concept cards and translations are data files, so new scenarios or languages are added without code changes. A translation parity test fails if any key is missing in Hindi or Telugu.
- **AI optional, offline first.** Every AI path has an offline fallback: built-in scenarios, an offline assessment and an offline mentor. Gemini calls try several models in turn on rate-limit or busy errors.
- **Persistence.** Progress is saved in the browser (localStorage) with versioned, validated storage, so a corrupt save resets safely.
- **Guardrail by design.** All investing content is labelled simulated, uses fictional outcomes and never names a real product or gives advice.

Design principles

- No shaming: a poor choice gets an explanation and a replay, not a "wrong" label.
- Accessible: keyboard and touch controls, focus management, visible focus rings, reduced-motion support, an audio-first mode, and layouts checked at 1280, 768 and 390 px.
- Local context: rupee amounts, chai carts, autos, UPI collect requests and KYC SMS scams that students actually see.

5. Testing and results

Test	What we ran	Result
Automated unit tests	161 Vitest tests in 17 files: Life Mode game rules, ledger and storage (50); 3D economy, missions, shops, traffic, pedestrian routes, movement, controls and results analysis (80); offline mentor, assessment, persona, progress and voice state (27); translation parity (4).	161 of 161 pass
Type check and build	`npm run check`: tsc over the JSDoc-typed domain module, all tests, production build.	Passes
3D mission walkthrough	Headless Chromium: walked to Ravi, took the delivery job, chose to save, then paid a phone repair from savings.	Cash, savings and mission step matched every choice
Pedestrian and traffic paths	Each pedestrian route simulated for two full loops; each car simulated for a full lap of the ring road.	No route touches a building; cars stay in lane
Offline mentor	"I spent 120 on lunch and got 500 from my aunt" with no AI key.	₹1,000 + ₹500 − ₹120 = ₹1,380, correct
Responsive and UI audit	Every route checked at 1280 and 390 px (Mentor also at 768), plus a written UI audit.	Fixed a 3D camera inside a building, 8 px sideways scroll on phones, an invisible clickable button in onboarding
AI resilience	Ran Learn, Assessment and Mentor with no key.	Offline fallbacks used, no AI call or console error

6. Limitations and next steps

- Progress is stored per browser; a small API and database would let NGOs track cohorts across devices.
- When an AI key is set it is used from the browser; a production deployment should route Gemini through a server-side proxy.
- Story text in Life Mode and Learn is still English; the interface is translated. Next: translate scenario content and add sign-language clips.
- Learner testing with five students without a finance background, observing which lessons they can explain back.